

# BASIC SECURITY ASSESSMENT

## WEEK 1

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**Role:** Cybersecurity Intern

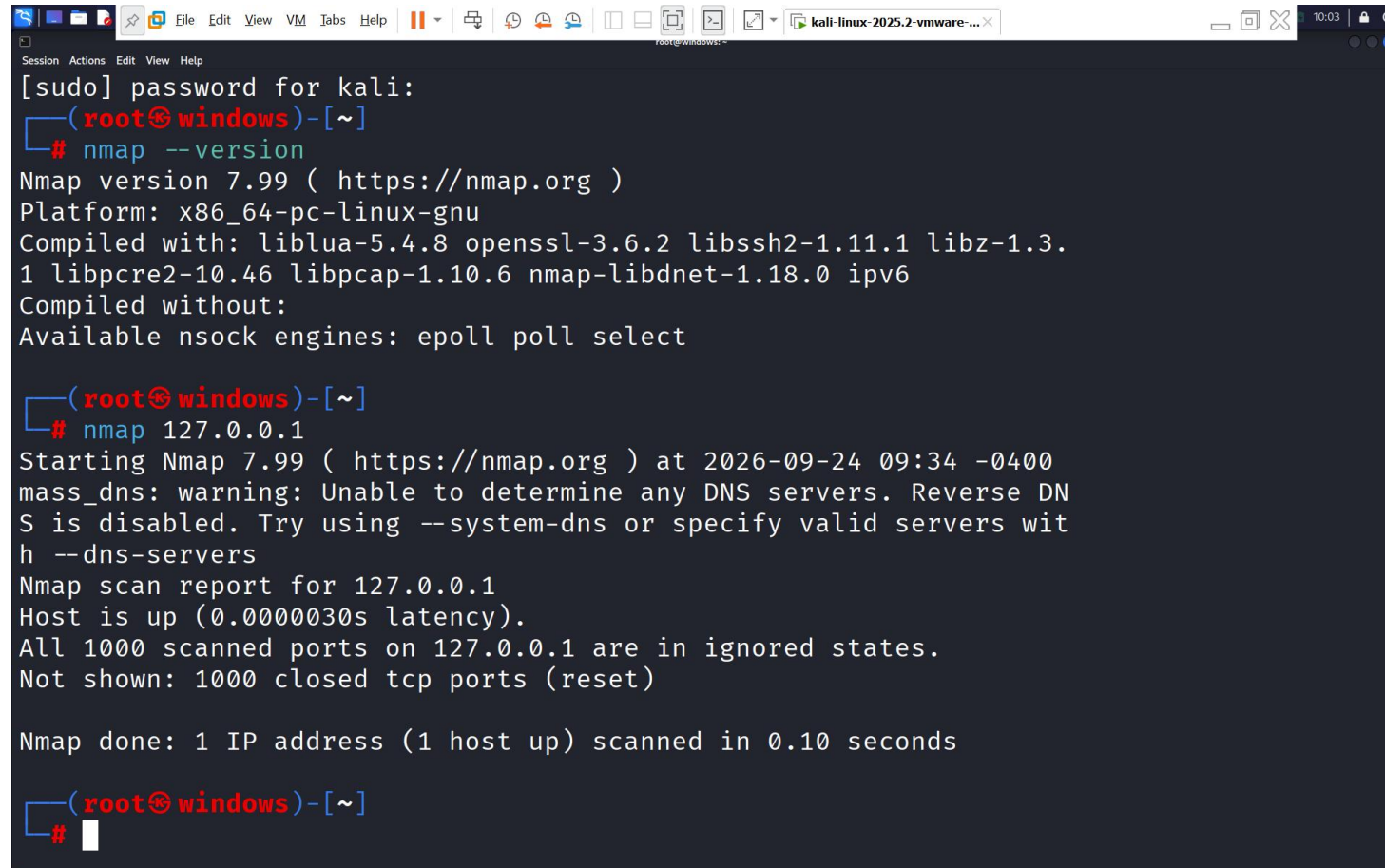
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# Nmap

- **Nmap (Network Mapper)** is a **network scanning tool** used to **discover devices, open ports, potential vulnerabilities and services on a network.**
- By default scans 1,000 common TCP ports.
- Host , port scan
- Tcp ,udp connect scan

# Nmap practical



```
Session Actions Edit View Help
[sudo] password for kali:
(root@windows)-[~]
# nmap --version
Nmap version 7.99 ( https://nmap.org )
Platform: x86_64-pc-linux-gnu
Compiled with: liblua-5.4.8 openssl-3.6.2 libssh2-1.11.1 libz-1.3.
1 libpcap-1.10.6 nmap-libdnet-1.18.0 ipv6
Compiled without:
Available nsock engines: epoll poll select

(root@windows)-[~]
# nmap 127.0.0.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-24 09:34 -0400
mass_dns: warning: Unable to determine any DNS servers. Reverse DN
S is disabled. Try using --system-dns or specify valid servers wit
h --dns-servers
Nmap scan report for 127.0.0.1
Host is up (0.0000030s latency).
All 1000 scanned ports on 127.0.0.1 are in ignored states.
Not shown: 1000 closed tcp ports (reset)

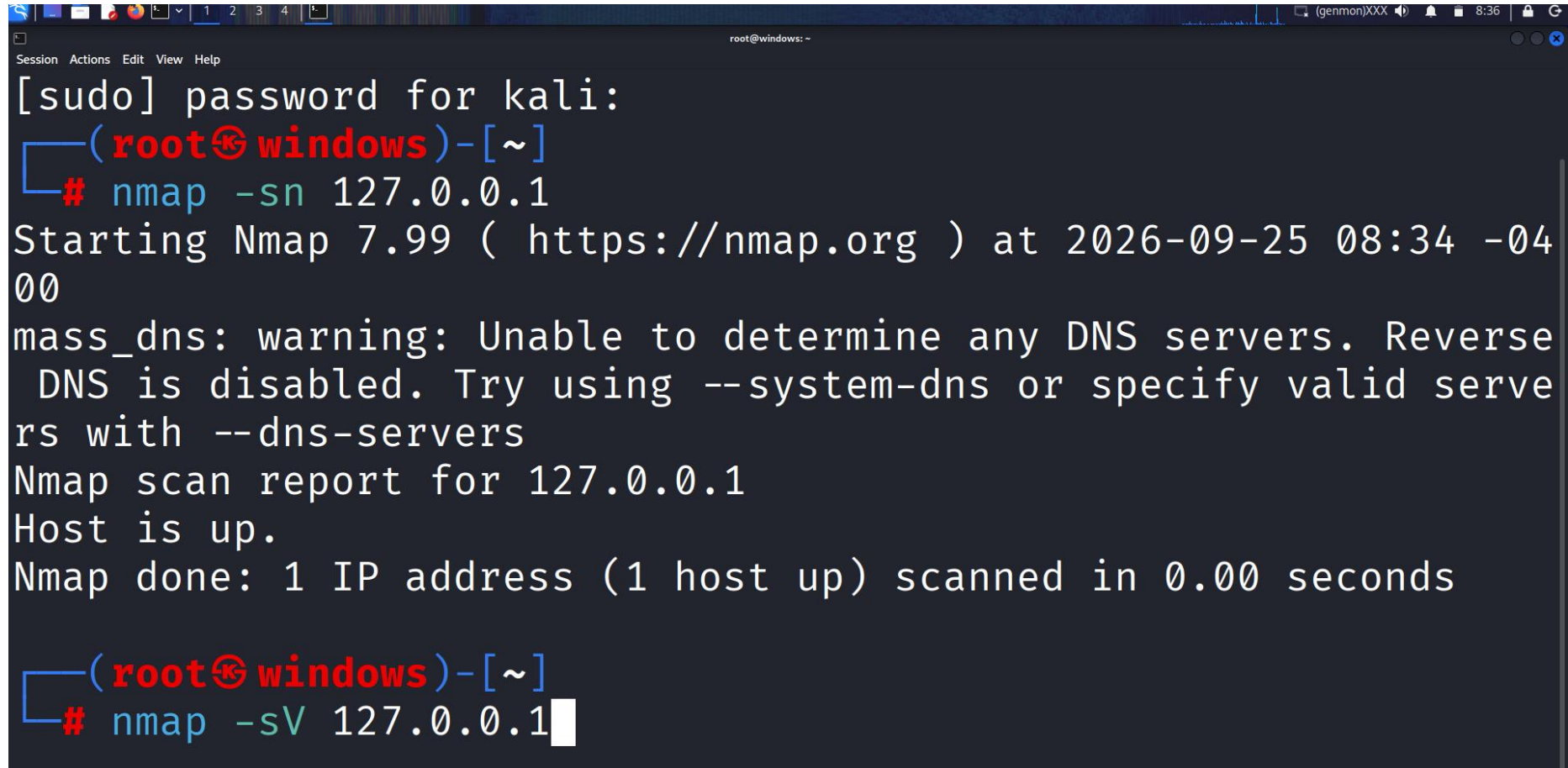
Nmap done: 1 IP address (1 host up) scanned in 0.10 seconds

(root@windows)-[~]
#
```

# Host Discovery

–sn : The -sn is used to detect live hosts without scanning for open ports.

**Syntax:** nmap –sn <ip address>

A terminal window titled 'root@windows: ~' showing the execution of the nmap command. The user enters '[sudo] password for kali:' and then '(root@windows)-[~]'. The command '# nmap -sn 127.0.0.1' is entered. The output shows 'Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-25 08:34 -0400', a warning about DNS, the scan report for 127.0.0.1 stating 'Host is up.', and the final summary 'Nmap done: 1 IP address (1 host up) scanned in 0.00 seconds'. The prompt returns to '(root@windows)-[~]' and the user enters '# nmap -sV 127.0.0.1' with a cursor at the end.

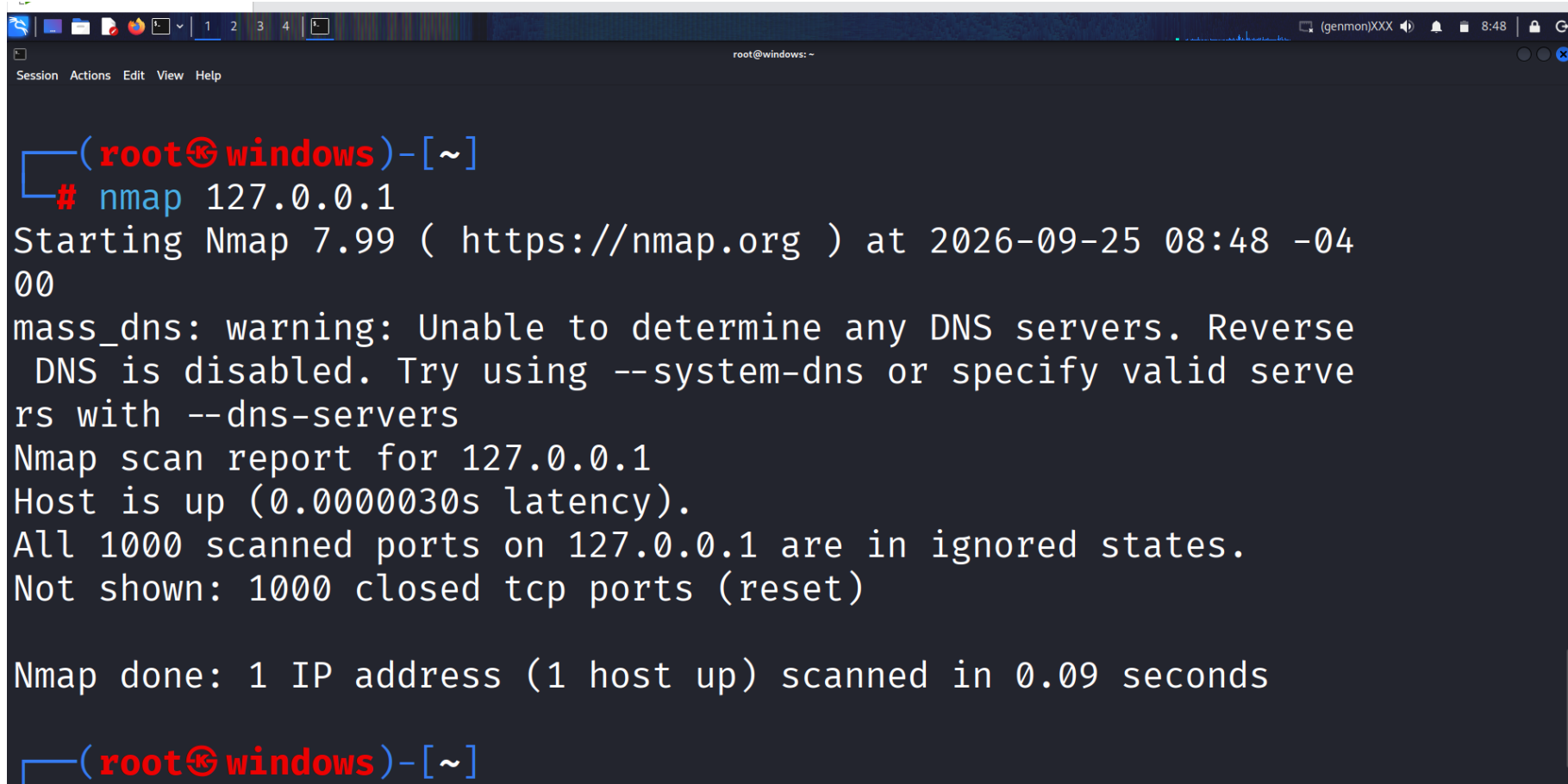
```
Session Actions Edit View Help
root@windows: ~
[sudo] password for kali:
(root@windows)-[~]
# nmap -sn 127.0.0.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-25 08:34 -0400
mass_dns: warning: Unable to determine any DNS servers. Reverse
DNS is disabled. Try using --system-dns or specify valid serve
rs with --dns-servers
Nmap scan report for 127.0.0.1
Host is up.
Nmap done: 1 IP address (1 host up) scanned in 0.00 seconds

(root@windows)-[~]
# nmap -sV 127.0.0.1
```

# Port scanning

To identify open ports on the target system.

During security assessments to find exposed services.

A terminal window titled 'root@windows: ~' with a menu bar (Session, Actions, Edit, View, Help) and a system tray (genmon)XXX, 8:48. The terminal shows a command prompt where the user runs 'nmap 127.0.0.1'. The output indicates that Nmap 7.99 is starting at 2026-09-25 08:48 -0400, with a warning about DNS servers. The scan report for 127.0.0.1 shows the host is up with a latency of 0.0000030s, and all 1000 scanned ports are in ignored states (reset). The scan is completed in 0.09 seconds.

```
(root@windows)-[~]
# nmap 127.0.0.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-25 08:48 -04
00
mass_dns: warning: Unable to determine any DNS servers. Reverse
DNS is disabled. Try using --system-dns or specify valid serve
rs with --dns-servers
Nmap scan report for 127.0.0.1
Host is up (0.0000030s latency).
All 1000 scanned ports on 127.0.0.1 are in ignored states.
Not shown: 1000 closed tcp ports (reset)

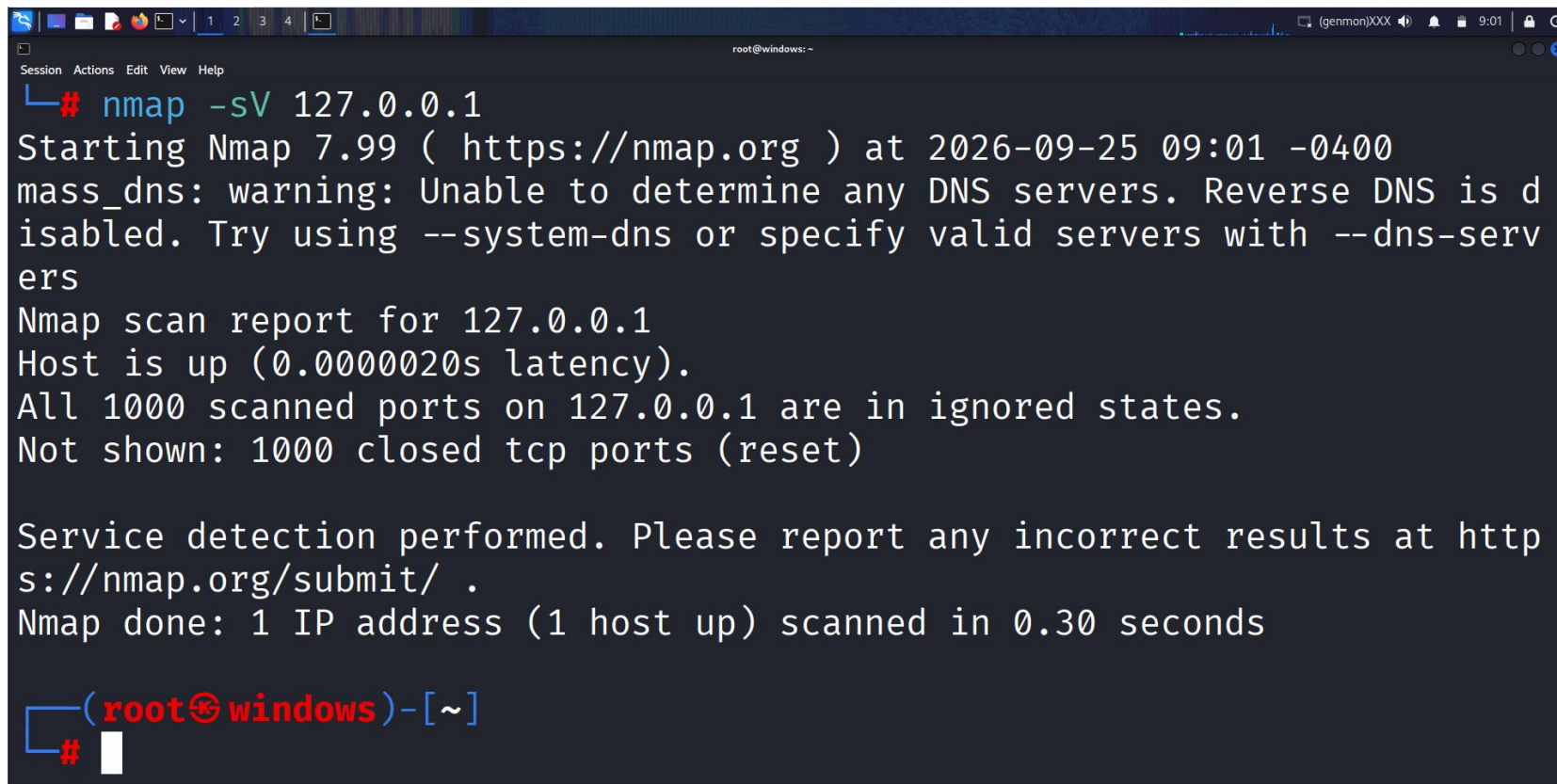
Nmap done: 1 IP address (1 host up) scanned in 0.09 seconds

(root@windows)-[~]
```

# SERVICE VERSION DETECTION

-sV: It helps identify **which services** are running on open ports and **what versions** they are using .

**Syntax:** nmap -sV <ip address>

A terminal window titled 'root@windows: ~' showing the execution of the nmap -sV 127.0.0.1 command. The output includes the Nmap version (7.99), a warning about DNS, a scan report for 127.0.0.1, and service detection results. The prompt at the bottom is '(root@windows)-[~]' with a cursor.

```
Session Actions Edit View Help
root@windows: ~
# nmap -sV 127.0.0.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-25 09:01 -0400
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 127.0.0.1
Host is up (0.0000020s latency).
All 1000 scanned ports on 127.0.0.1 are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 0.30 seconds

(root@windows)-[~]
#
```

# OS Detection

To identify the operating system of the target host.



```
root@windows: ~  
Session Actions Edit View Help  
(root@windows)-[~]  
# nmap -O 127.0.0.1  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-25 09:03 -0400  
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers  
Nmap scan report for 127.0.0.1  
Host is up (0.000081s latency).  
All 1000 scanned ports on 127.0.0.1 are in ignored states.  
Not shown: 1000 closed tcp ports (reset)  
Too many fingerprints match this host to give specific OS details  
Network Distance: 0 hops  
  
OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 1.57 seconds  
  
(root@windows)-[~]  
#
```

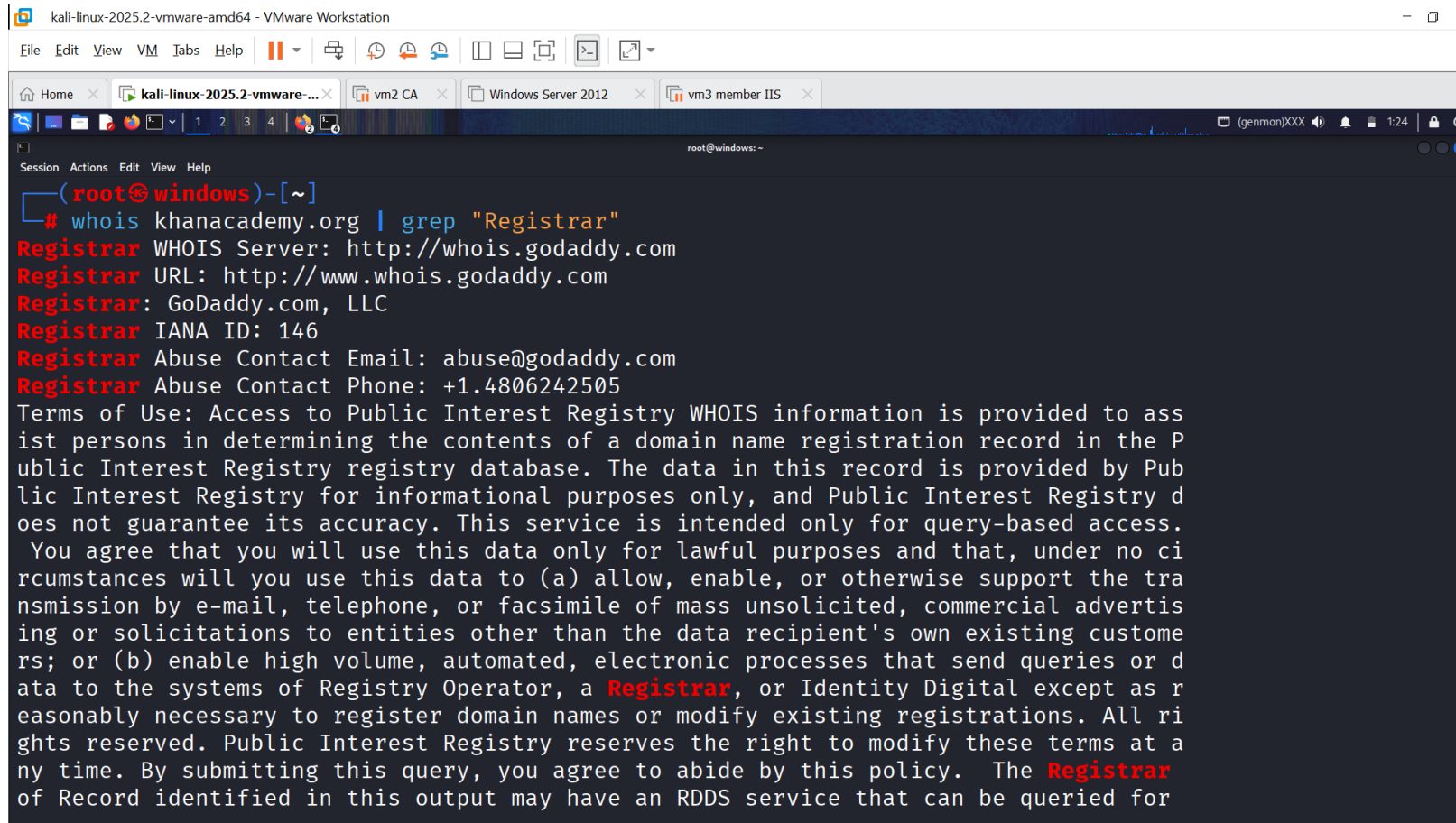
# WHOIS

- A WHOIS lookup is the process of querying a WHOIS database to obtain registration information about a domain name or IP address.
- It helps identify ownership and administrative details of internet resources.

## **Information Provided by WHOIS:**

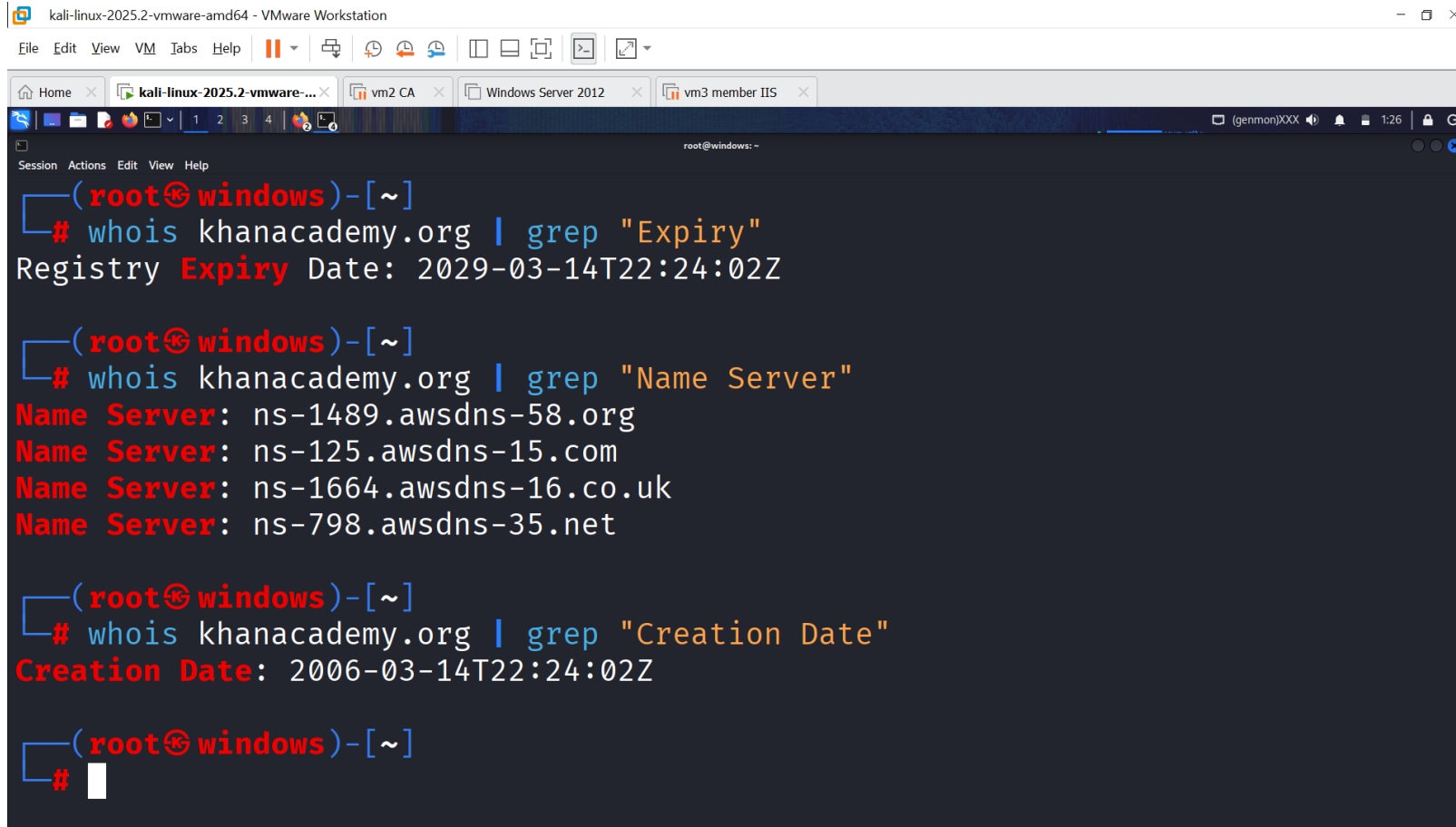
- Domain Name
- Registrar
- Registration Date
- Expiration Date
- Name Servers
- Registrant Information like Organization name, Country Contact email, Administrative contact, Technical contact
- Many domains now use privacy protection services, so personal details may be hidden.

Whois cmd – is use to check the domain owner, registration/expiry dates, registrar, nameservers, and domain status to know who owns it and how the domain is managed



```
kali-linux-2025.2-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.2-vmware-... vm2 CA Windows Server 2012 vm3 member IIS
root@windows: ~
Session Actions Edit View Help
(root@windows)-[~]
# whois khanacademy.org | grep "Registrar"
Registrar WHOIS Server: http://whois.godaddy.com
Registrar URL: http://www.whois.godaddy.com
Registrar: GoDaddy.com, LLC
Registrar IANA ID: 146
Registrar Abuse Contact Email: abuse@godaddy.com
Registrar Abuse Contact Phone: +1.4806242505
Terms of Use: Access to Public Interest Registry WHOIS information is provided to assist persons in determining the contents of a domain name registration record in the Public Interest Registry registry database. The data in this record is provided by Public Interest Registry for informational purposes only, and Public Interest Registry does not guarantee its accuracy. This service is intended only for query-based access. You agree that you will use this data only for lawful purposes and that, under no circumstances will you use this data to (a) allow, enable, or otherwise support the transmission by e-mail, telephone, or facsimile of mass unsolicited, commercial advertising or solicitations to entities other than the data recipient's own existing customers; or (b) enable high volume, automated, electronic processes that send queries or data to the systems of Registry Operator, a Registrar, or Identity Digital except as reasonably necessary to register domain names or modify existing registrations. All rights reserved. Public Interest Registry reserves the right to modify these terms at any time. By submitting this query, you agree to abide by this policy. The Registrar of Record identified in this output may have an RDDS service that can be queried for
```

# Whois practical



The screenshot shows a Kali Linux terminal window titled "kali-linux-2025.2-vmware-amd64 - VMware Workstation". The terminal is running a series of whois commands on the domain "khanacademy.org". The first command filters for the "Expiry" date, showing it as 2029-03-14T22:24:02Z. The second command filters for "Name Server", listing four servers: ns-1489.awsdns-58.org, ns-125.awsdns-15.com, ns-1664.awsdns-16.co.uk, and ns-798.awsdns-35.net. The third command filters for the "Creation Date", showing it as 2006-03-14T22:24:02Z. The terminal prompt is root@windows: ~.

```
(root@windows)-[~]
# whois khanacademy.org | grep "Expiry"
Registry Expiry Date: 2029-03-14T22:24:02Z

(root@windows)-[~]
# whois khanacademy.org | grep "Name Server"
Name Server: ns-1489.awsdns-58.org
Name Server: ns-125.awsdns-15.com
Name Server: ns-1664.awsdns-16.co.uk
Name Server: ns-798.awsdns-35.net

(root@windows)-[~]
# whois khanacademy.org | grep "Creation Date"
Creation Date: 2006-03-14T22:24:02Z

(root@windows)-[~]
#
```

# Shodan

- **Shodan** (often called the "**Search Engine for Internet-Connected Devices**") is a specialized search engine that indexes devices connected to the Internet instead of web pages.

## **Shodan searches:**

- Servers
- Routers
- Switches
- Firewalls
- Printers
- Industrial Control Systems (ICS)
- Databases
- Web servers

# Shodan

## Search by product:

Shodan

Maps

Images

Monitor

Developer

More...

 SHODAN

Explore

Downloads

Pricing [↗](#)

product:Apache

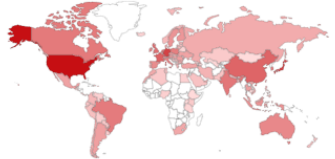


Account

TOTAL RESULTS

11,075,123

TOP COUNTRIES



|               |           |
|---------------|-----------|
| United States | 3,383,893 |
| Japan         | 1,168,703 |
| Germany       | 1,139,253 |
| France        | 495,557   |
| China         | 455,079   |

More...

TOP PORTS

|     |           |
|-----|-----------|
| 80  | 5,834,924 |
| 443 | 4,124,363 |

 View Report

 Browse Images

 View on Map

 Advanced Search

Product Spotlight: A native desktop app to access Shodan using your API key. Fast and optimized for power users. Check out [Shodan Terminal](#)

 **Apache Tomcat/9.0.71** [↗](#)

105.159.159.179  
[IAM](#)  
 Morocco, Casablanca

HTTP/1.1 200  
Content-Type: text/html; charset=UTF-8  
Transfer-Encoding: chunked  
Date: Sat, 04 Jul 2026 08:51:20 GMT

2026-07-04T08:56:27.789920

**ERROR: The requested URL could not be retrieved** [↗](#)

153.76.127.184  
[NCR Corporation](#)  
 United States, Peachtree City

HTTP/1.1 400 Bad Request  
Server: Apache/2.4.41 (Ubuntu) Linux  
Mime-Version: 1.0  
Date: Sat, 04 Jul 2026 08:50:30 GMT  
Content-Type: text/html; charset=utf-8  
Content-Length: 3424  
Connection: close

2026-07-04T08:55:38.058166

**47.96.155.182** [↗](#)

[Aliyun Computing Co., LTD](#)  
 China, Hangzhou

HTTP/1.1 200 OK  
Server: Apache-Coyote/1.1  
Content-Type: text/html; charset=UTF-8  
Transfer-Encoding: chunked

2026-07-04T08:55:21.344744

 **Apache Tomcat** [↗](#)

2026-07-04T08:54:30.057564

# Search by IP address:

 SHODAN

Explore

Downloads

Pricing ↗

ip:8.8.8.8



Account

TOTAL RESULTS

2

TOP PORTS

|     |   |
|-----|---|
| 53  | 1 |
| 443 | 1 |

 View Report

 View on Map

 Advanced Search

Product Spotlight: A native desktop app to access Shodan using your API key. Fast and optimized for power users. Check out [Shodan Terminal](#)

**8.8.8.8**

dns.google  
Google LLC  
United States, Mountain View

Recursion: enabled

2026-07-03T22:02:34.127541

 **Google Public DNS** ↗

8.8.8.8  
dns.google  
cloud.e-core.com  
Google LLC  
United States, Mountain View

 **SSL Certificate**

Issued By:  
|- Common Name:  
**WR2**  
|- Organization:  
**Google Trust Services**  
Issued To:  
|- Common Name:  
**dns.google**  
Supported SSL Versions:  
**TLSv1.2, TLSv1.3**

HTTP/1.1 200 OK  
Content-Security-Policy: object-src 'none';base-uri 'self';script-src 'nonce-9p9iUhXAkdn90f1QGnTfdg' 'str  
Strict-Transport-Secur...



2026-07-04T08:32:14.421092